



NOVA SCHOOL OF
SCIENCE & TECHNOLOGY

MIEEC/MIEMN

2025 / 2026

Course : **Electrónica de Rádio Frequência**

Low Noise Amplifier – Part 2

Goal:

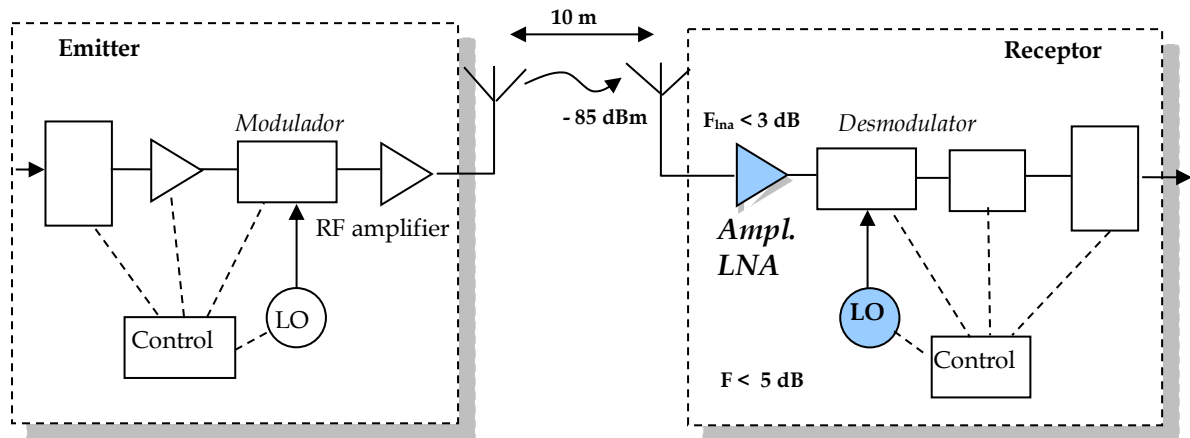
Analysis and design of a CMOS LNA.

<http://moodle.fct.unl.pt>

Version 1.0
Date: 12th May 2025
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Motivation

Modern telecommunications systems use high-frequencies with high data-rates. In the receiver path the critical block is the LNA, since, it needs to deal with very low amplitude signals. The LNA has stringent specifications in terms of gain, noise, IM3, P_{1dB} ... In this Project it is expected that the students will understand in detail all these aspects designing an LNA for a ISM (industrial, scientific and medical) band.



Specifications

The LNA should fulfill the following specifications:

- $S_{11} < -10 \text{ dB}$;
- $S_{22} < -10 \text{ dB}$;
- $F < 3 \text{ dB}$;
- $\text{Gain} > 10 \text{ dB}$ (= MAG 1st part);
- 0.1 - 2 GHz with 350 nm CMOS and 65nm (5GHz) and 45 nm (10 GHz);

Project assignment

Group	Transistor 350 nm CMOS	Group	Transistor 350 nm CMOS
P1G1	VDD = 3 V n = 1.75	P2G1	VDD = 2.5 V n = 1.75
P1G2	VDD = 3 V n = 2	P2G2	VDD = 2.5 V n = 2
P1G3	VDD = 3 V n = 2.25	P2G3	VDD = 2.5 V n = 2.25
P1G4	VDD = 3 V n = 2.5	P2G4	VDD = 2.5 V n = 2.5
P1G5	VDD = 3 V n = 2.75	P2G5	VDD = 2.5 V n = 2.75
P1G6	VDD = 3 V n = 3	P2G6	VDD = 2.5 V n = 3
P1G7	VDD = 3 V n = 3.25	P2G7	VDD = 2.5 V n = 3.25

Optional: a) technology 65nm (cadence) VDD=1.2 V, b) technology of 45nm (LTSPICE) VDD = 1 V

Components

➤ Transistors

Using 350 nm and optional: 45 nm CMOS technology in simulator LTSPICE and 65nm (cadence in Lab)

Design procedure

1- Project:

LNA

- a. Design the LNA using the circuit topology combined CG + CS stages.
 - i. Design Transistor Bias and sizing for CG and CS (each stage gain = MAG or > 10 dB).
 - ii. Determine the gain and NF.
 - iii. Implement the ratio n between CG and CS gm.
 - iv. Implement the LNA with minimum L (350nm) and compare with 130nm (optional) and 45 nm technology.
 - v. Design the output buffer for output Z0 match.
- b. Simulation:
 - i. Validate LNA design using Spice.
 - ii. Simulate ratio n between CG and CS gm and check the differences in Gain, NF and Power consumption.
 - iii. Compare the results between different CMOS technologies.

2- Report

- a. The report should have:

- i. Detailed justification about the options done during the project.
 - ii. Circuit schematics, simulations files, plot, simulation results.
- b. Structure:
 - i. First page with the group number and students identification.
 - ii. Introduction
 - 1. Objective
 - 2. Motivation
 - iii. Main Body
 - 1. Design
 - 2. Simulations
 - 3. Final circuit
 - 4. Analysis of Results
 - 5. Comparison between bipolar and CMOS design.
 - iv. Conclusions

3- Delivery

- a. Date: 12-06-2025
- b. Place: upload in moodle

Contacts

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